

Education

Virginia Tech

Master of Science in Computer Engineering

Aug 2024 – May 2026

Blacksburg, United States

- **Relevant Coursework:** Computer Vision, Advanced Machine Learning, Stochastic Signals and Systems, Non-Linearity and Predictability, Quantum Engineering, Computer Architecture, Statistics, Network Architecture and Protocols

Experience

Founding Engineer

May 2025 – Present

Cinemind

- Designed and launched an **AI-powered**, LLM-integrated IP coverage platform for film studios, reducing evaluation time from days to minutes and delivering professional-quality reports used by producers for greenlighting decisions, saving clients significant time and money.
- Enabled **global, executive-friendly** access by building a fully functional web interface with scalable multi-IP processing, allowing simultaneous analysis of multiple screenplays and aligning features with studio workflows for maximum adoption.
- Architected and deployed a secure, cost-efficient **AWS infrastructure** with industry-grade security policies to protect confidential IP, and integrated **Stripe** for seamless subscription and one-time payment processing, streamlining onboarding and billing. *Tech stack:* AWS (EC2, S3, IAM), Stripe API, OpenAI, Gemini, Docker.

Graduate Research Assistant

Oct 2024 – May 2025

Virginia Tech

Blacksburg, VA

- Developed a **state-space model of fear propagation** during disasters, integrating **cooperation, power, fake news, and risk**; derived a **two-time-scale nonlinear system** and proved stability via **Lyapunov analysis**.
- Built a **decentralized multi-agent control framework** combining **MADDPG, MPC, and game-theoretic resource allocation**; trained on **CUDA GPUs**, achieving **30% lower peak panic** and **22% faster response** in **10k+ agent simulations**.
- Presented “Fear Dynamics and Multi-Agent Control” at **NESCW’25 (Columbia University)**, showcasing **control theory, reinforcement learning, and game theory for disaster-response optimization**.

Machine Learning Engineer Intern

Nov 2023 – Jul 2024

Devot AI

Bengaluru, India

- Evaluated and annotated outputs from **OpenAI** models for 2k+ prompts across coding, math, reasoning, and search tasks, providing corrective feedback to improve accuracy, reduce hallucinations, and enhance security-focused responses by 15%.
- Built interactive dashboards for media analytics, improving campaign returns by 10–15% through data-driven insights.
- Built and maintained scalable, containerized data pipelines in **Python** and **Docker** on **AWS**, supporting scheduled and on-demand workloads.

Web Developer Intern

Oct 2021 – Jan 2022

Ankureto Store

Pune, India

- Developed an e-commerce website, **boosting online sales by 25% and user engagement by 40%** using React, HTML, CSS, JavaScript, and GitHub.
- Optimized site performance, reducing page load times by 50% and **integrating front-end and back-end components for a 20% performance gain**.

Projects

Veena.ai | Live Demo

Apr 2025 – May 2025

- Built a Perplexity-style browser chatbot with real-time web search + retrieval-augmented generation on **Qwen-3 32B**; streams cited answers and handles conversational follow-ups.
- Unified **OpenAI, Anthropic, and Groq** behind a common API with focus modes (web/media/docs); and **production deployment on Vercel**, onboarding an initial user base via private links, and submitted to **YC Startup School S25**.

GraphDocuMind | Github

Feb 2025 – Mar 2025

- Built a **retrieval-augmented chatbot** that delivers instant, cited answers from course notes; organized **10k+ facts** into a **Neo4j graph database**, raising top-3 accuracy to **93%**.
- Automated ingestion of slide decks & PDFs via **LlamaParse**; chunked & embedded content and served responses from a locally hosted **Llama3** model for full privacy.
- Exposed the graph through **GraphQL APIs** on top of Neo4j, enabling flexible, structured queries and seamless front-end integration.

ChalBey | Live Website

Jan 2025 – Mar 2025

- Built to end long waits for car-less international students: lets classmates share rides, turning empty seats into new friendships.
- Full-stack app (Node/Express + PostgreSQL) deployed on AWS/GCP with Docker containers; real-time geospatial matching paired riders in less than 2s; facilitated 100+ shared trips during beta.

Lagori Robot | Abu Robocon | Video Link

Aug 2021 – Jun 2022

- Represented my college at **Abu Robocon 2022** (theme “Lagori”), building two cooperative robots that used **YOLO** vision (95 % accuracy) to knock down and restack targets.
- Programmed and integrated **Raspberry Pi, Arduino Uno, and OpenMV** modules; calculated forces and torques to size motors and select electronics, ensuring reliable movement and throwing during matches.

Technical Skills

Languages: Python, C, C++, JavaScript, Java, TypeScript, SQL, MATLAB, HTML/CSS (Object-Oriented Programming) **Frameworks / Libraries:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, LangChain, React.js, Node.js, CUDA **Developer & Cloud Tools:** VS Code, Cursor/Windsurf, Git/GitHub, Docker, AWS, Azure, GCP, Vercel, Railway **Databases:** PostgreSQL, Snowflake, MySQL, MongoDB **Domains:** Computer Science, Software Engineering, Distributed Systems, Computer Networks, Machine Learning, Deep Learning, Reinforcement Learning, Computer Vision, Data Science, Natural Language Processing (NLP), Robotics, Full-Stack Development **Specialized Expertise:** Large Language Models (LLMs), Vision Transformers, RESTful API Development, Geospatial Indexing (PostGIS), Model Predictive Control (MPC), Multi-Agent RL (MADDPG)